Online Pipeline Weld Defect Detection for Magnetic Flux Leakage Inspection System via Lightweight Rotated Network

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***Abstract*—Developing an online pipeline magnetic flux leakage (MFL) weld defect detection method deployed in industrial sites is essential to expedite the localization and repair of defects. However, most existing methods suffer from large computational burdens, which hinder their deployment on resource-limited industrial personal computers (IPCs) in sites. Additionally, weld defects in MFL signals exhibit irregular contours, and there exist strong couplings between welds and defects. To tackle these limitations, this study proposes an online weld defect detection method for MFL inspection systems via a lightweight rotated network (LRNet). Initially, an efficient feature encoder is devised to mine distinguishable representations of weld defects to clarify the coupled relations between welds and defects. Subsequently, a dual-path feature aggregation is established to enrich the semantic and localization information of the extracted features through top-down and bottom-up subnetworks. Finally, a refined rotated boundary decision is designed to accurately characterize and predict the irregular contours of weld defects. The three tailored strategies follow the lightweight design principle aimed at alleviating the model's computational burden. Experimental results on MFL data from experimental sites and the real world demonstrate that LRNet improves accuracy by approximately 10% under comparable computational costs, which facilitates its deployment in IPCs.**

***Index Terms*—Coupled relations, irregular contours, lightweight rotated network (LRNet), magnetic flux leakage (MFL), online weld defect detection, pipeline.**

I. Introduction

PIPELINE transportation is the primary mode of energy transportation in modern society. However, welds, as the vulnerable link in pipeline connections, are prone to failures due to the interaction of internal transportation media and external environmental factors [1]. The stress concentration caused by weld defects will significantly shorten the service life of the pipeline or even induce pipeline fracture, potentially leading to safety incidents and casualties. In view of this, accurate weld defect detection is essential to ensure the safe operation of oil and gas pipelines [2].

Magnetic flux leakage (MFL) has emerged as a primary nondestructive testing technique for various metallic materials, such as wire ropes [3], [4], steel plates [5], and pipelines [6], due to its reliability, automation, and environmental friendliness. For oil and gas pipelines, the existing inspection process utilizes MFL robots to complete the inspection operation, followed by offline data analysis and defect identification by engineers. This offline inspection process requires a lot of manpower and computing power, and it takes at least 30 days of lengthy processing to give inspection results, which makes it difficult to report pipeline safety issues in a timely manner, presenting a huge potential risk. However, pipeline owners urgently need to know about serious risks immediately after the inspection is completed. Current identification methods do not meet the practical requirements of this online application. Therefore, there is an urgent need to develop a lightweight online detection method to address the technical limitations between the practical needs and the existing methods.

The limitations of existing methods are primarily reflected in two aspects. On the one hand, accurate detection models require substantial computational power, which cannot be implemented in industrial sites. Specifically, researchers often achieve superior performance by designing complex model structures [7], [8], but this leads to large computational overheads and limited inference speeds, which are difficult to deploy in industrial

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personal computers with limited resources. Additionally, with the improvement of sensor accuracy, the amount of data collected after long-distance pipeline inspection can reach several terabytes, so the limited inference speed is inadequate for obtaining timely inspection results [9].

On the other hand, the accuracy of lightweight models does not meet the practical engineering requirements. Lightweight techniques aim to compress the model parameters and computational complexity while striving to maintain the model's accuracy, which makes them a promising tool for realizing online detection. Recently, researchers have deeply investigated lightweight defect detection methods and yielded fruitful achievements. For instance, Liu et al. [10] proposed a lightweight and faster you only look once (LF-YOLO) approach for weld defect detection using convolutional neural networks (CNNs). Zhu et al. [11] introduced a lightweight multiscale attention semantic segmentation algorithm with high precision and efficiency to detect laser weld defects. Liang et al. [12] designed a lightweight efficient detection network (LED-Net) using deep dynamic convolution as the fundamental structure. Liu et al. [13] developed a pyramid feature CNN method specifically for rail surface defect detection. Wang et al. [14] incorporated modified cross-stage-partial-darknet (MC) and coordinated attention (CA) into you only look once X (YOLOX) and built MC-CA-YOLOX for effective detection of welding defects in the printed circuit board (PCB). Wu et al. [15] performed region of interest (ROI) selection on the original defect image and then constructed a lightweight CNN classification network to determine the ROI category. Additionally, leveraging the knowledge distillation technique, [16] and [17] proposed novel approaches for industrial defect detection by integrating Centernet and reparameterization methods, respectively. However, although these studies have achieved elegant lightweight structures, their promised performance may be degraded when faced with MFL weld defect detection due to the following technical limitations: 1) the strong coupled relations between welds and defects hinder the model from extracting distinguishable features; and 2) the horizontal bounding box annotation used by most existing methods is insufficient to describe the irregular contour of weld defects under the MFL signal.

To address the first limitation, an efficient feature encoder is constructed to mine distinguishable features of weld defects using three core modules: multiview context fusion (MVCF), efficient cross-feature interaction (ECFI), and dynamic gated channel attention (DGCA). Specifically, MVCF improves the preservation of fine-grained features during compression and concatenation, ECFI facilitates the fusion of critical features, and DGCA enhances the inter-channel feature interaction, which enables the three modules to work together to clarify the coupled relations between welds and defects. In addition, to further enhance the detection performance, a dual-path feature aggregation structure is constructed by embedding the three tailored modules, where the top-down and bottom-up subnetworks can enrich the semantic representations and localization information in the multiscale features, thus further enhancing the discrimination of the features.

To overcome the second limitation, we investigated the utilization of rotated bounding boxes to characterize irregular contours of weld defects under MFL signals, which provides a more appropriate annotation form that can mitigate background interference. However, the introduction of angles entails boundary discontinuity and square-like issues. [18]. Additionally, welds are slender objects with extreme aspect ratios, which greatly affects the detector's performance. To tackle these issues, this article proposes a refined rotated boundary decision strategy that consists of two components. On one hand, a fast decoupled head (FD-Head) is constructed by incorporating MVCF to fully exploit the strong nonlinear relation of MFL signals. On the other hand, distributed consistency (DC) and aspect ratio (AR) losses are designed, where DC loss achieves robust boundary regression and AR loss guides the model to capture the weld's slender features.

In summary, this article proposes an online intelligent weld defect detection method called lightweight rotated network (LRNet). Specifically, an efficient feature encoder and dual-path feature aggregation are constructed to extract and aggregate distinguishable representations of welds and defects. A refined rotated boundary decision strategy is customized to realize accurate contour regression of weld defects. To the best of our knowledge, this article is the first study to explore the use of rotation-based detection methods to complete the online detection of pipeline weld defects under MFL signals.

The contributions of this article are summarized as follows.

1) An efficient feature encoder is constructed using three developed modules (MVCF, ECFI, and DGCA) to extract recognizable representations of weld defects from coarse to fine and to distinguish the coupled relations between welds and defects.
2) A dual-path feature aggregation method is designed to fully aggregate the multiscale features of weld defects, in which the top-down and bottom-up subnetworks enrich the abstract semantics and shallow localization information of features at different scales, and the embedding of the three core modules strengthens the integration of the high-level representation of weld defects.
3) A refined rotated boundary decision is proposed for finer localization of weld defects, where FD-Head enhances the learning of the MFL nonlinear mechanism, the DC loss robustly regresses coordinates and angles, and AR loss guides the model to focus on the weld's slender features.

The remainder of this article is organized as follows: Section II elaborates on the MFL online inspection system and the problems to be addressed. In Section III, the proposed method is presented in detail. Section IV conducts experiments and discusses the obtained results. Finally, Section V concludes the article and outlines future research directions.

II. Online Pipeline MFL Inspection System and Problem Statement

A. Online Pipeline MFL Inspection System

The online pipeline MFL inspection system, as shown in Fig. 1, is divided into five steps. *Step 1*. For the pipe to be

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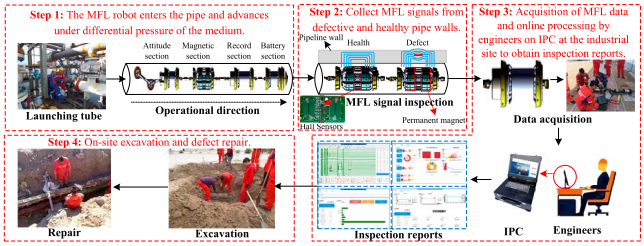


Fig. 1. Intelligent online pipeline MFL inspection systems.

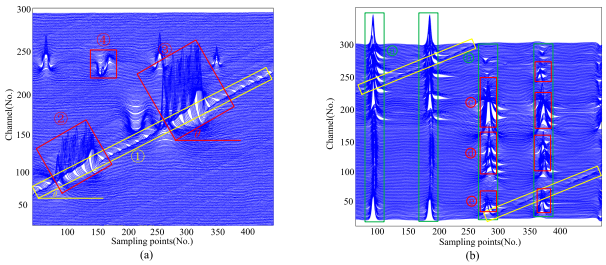


Fig. 2. Actual MFL signals of weld defects. (a) Coupled spiral weld and defect. (b) Coupled girth weld and defect.

tested after pigging, the MFL robot enters its interior through the launching tube and advances under the pressure difference of the medium. *Step 2.* During the advancement process, the magnetic section equipped with multiple pairs of permanent magnets magnetizes the pipe wall until it reaches magnetic saturation. When there is a defect in the pipe wall, the magnetic field lines will change their path due to the change in permeability and form a leakage flux at the defect, which can be captured as MFL signals by the Hall sensors carried by the robot. *Step 3.* Engineers download the collected MFL data in the receiving tube and process it online using an industrial personal computer (IPC) to obtain an inspection report. *Step 4.* Professional pipeline repairers carry out on-site excavation and defect repair in a timely manner based on the inspection report to ensure stable operation of the pipeline. In contrast to the offline inspection process, the above process is completed online on the IPC of the industrial site to timely know the serious risks of the pipeline.

B. Problem Statement

Compared with normal defects in the pipe wall, weld defects located in vulnerable areas of the pipeline are susceptible to failure and seriously threaten the operational safety of the pipeline. Due to the high risk of weld defects, efficient and accurate detection of weld defects is a crucial data processing link in an online pipeline MFL inspection system. Fig. 2 depicts the obtained MFL signals of weld defects. From Fig. 2, it can be seen that there exists a regional coupling between the spiral weld (①) and the defects (②, ③) and between the girth weld (ⓑ) and the defects (ⓒ, ⓓ, ⓔ). On the other hand, there exists distributional coupling between the girth weld (ⓐ) and the defects (ⓒ, ⓓ, ⓔ). Consequently, the interaction of regional and distributional coupling weakens the discriminative feature extraction of the model. In addition, compared with normal defects (④), spiral welds and defects possess irregular shapes with angles, which complicates defect identification. To address these issues, this article proposes an LRNet that aims to effectively extract discriminative features of weld defects to clarify the coupled relations between welds and defects and accurately predict the contours of weld defects to improve the timeliness of defect repair.

III. Proposed Method

A. Overall Framework of LRNet

The overall framework of LRNet is illustrated in Fig. 3, which comprises four parts: sample generation, efficient feature encoder, dual-path feature aggregation, and refined rotated boundary decision. The sample generation aims to generate high-quality training samples from raw signals through signal preprocessing and conversion. In the efficient feature encoder, three lightweight core modules are devised to efficiently encode the critical features of weld defects from macroscopic to microscopic scales. Dual-path feature aggregation combines top-down and bottom-up subnetworks with core design modules to fully aggregate the multiscale representation of weld defects. Finally, the proposed DC and AR losses are utilized to assist FD-Head accurately regress the boundaries of weld defects in the refined rotated boundary decision.

B. Sample Generation

The purpose of sample generation is to alleviate the adverse impact of the raw MFL signal and generate high-quality training samples, which involves two steps: signal preprocessing and conversion.

1) Signal Preprocessing $P(\bullet)$: The data preprocessing includes baseline correction, interpolation, and filtering [8].

Step 1: The baseline correction is conducted to establish consistent reference values for data with similar attributes obtained from different sensors, computed as follows:

$$S'(i,j) = S(i,j) - \frac{1}{N}\sum_{i=1}^{N} S(i,j) + \frac{1}{MN}\sum_{j=1}^{M}\sum_{i=1}^{N} S(i,j) \tag{1}$$

where $S(i,j)$ and $S'(i,j)$ denote the original and corrected signals, respectively; M and N are the number of Hall sensors and the sampling points.

Step 2: Due to the inherent uncertainty in signal capture, cubic spline interpolation is used to assure continuity and dependability. Furthermore, Gaussian filtering is employed to reduce noise and generate a pure signal $\tilde{S}$ defined as

$$\tilde{S}(i,j) = \Theta(S'(i,j)) * \frac{1}{2\pi\sigma^2} e^{-\frac{i^2+j^2}{2\sigma^2}} \tag{2}$$

where $\Theta(\bullet)$ represents interpolation, $*$ represents convolution, and σ represents the standard deviation of the Gaussian kernel.

2) Signal Conversion $C(\bullet)$: The signal conversion aims to increase the discernibility of defect signals. For ease of conversion, $\tilde{S}$ is first split into $\tilde{S}'$ [19]. Then, $\tilde{S}'$ can be mapped to

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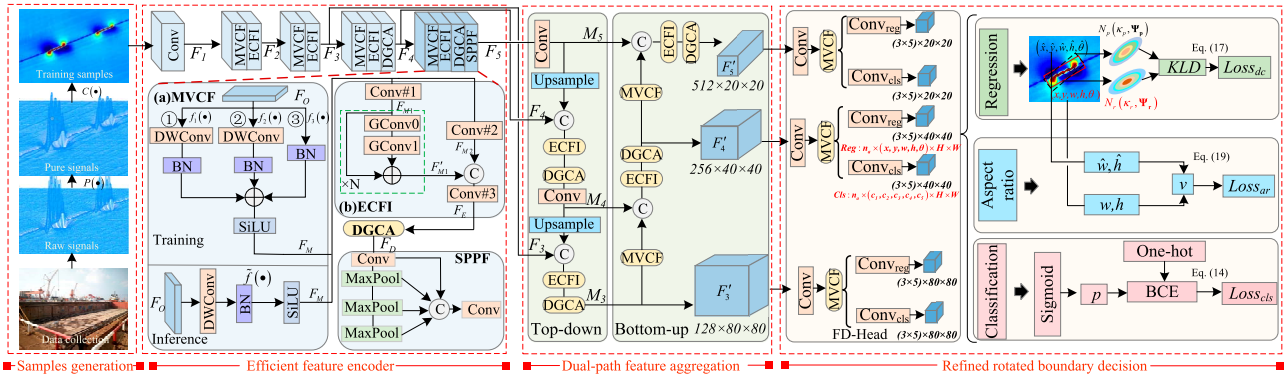


Fig. 3. Overall framework of the proposed lightweight rotated network. In FD-Head, n_a denotes the number of anchors assigned to each grid, (x, y, w, h, θ) represents the regressed RBB coordinates, and $(c_1, c_2, c_3, c_4, c_5)$ represents the confidence score of the i-th class.

the pseudo-color image $Y \in \mathbb{R}^{H\times W\times 3}$ using the preset color conversion matrix $M \in \mathbb{R}^{\delta\times 3}$, which is denoted by

$$Y = M\left[\left[\delta * \frac{\tilde{S}' - \min}{\max - \min}\right](i,j),:\right] \tag{3}$$

where δ denotes the image resolution set to 1024, and $\lceil\bullet\rceil$ represents rounding up to the nearest integer.

C. Efficient Feature Encoder

Given the coupled relations of welds and defects in MFL signals and the industry's requirement for real-time inference speed, three core modules, i.e., MVCF, ECFI, and DGCA are designed with a lightweight idea to improve the model's nonlinear learning capability in distinguishing features between weld and defect.

1) Multiview Context Fusion (MVCF): Model downsampling may lead to the loss of key feature details, which makes it difficult for the model to obtain discriminative features of MFL weld defects. To address this issue, an MVCF is designed to preserve defect-specific fine-grained features from multiple perspectives during feature compression. This module decouples training and inference, which improves model performance without increasing inference costs [20]. Furthermore, depthwise convolutions (DWConv) effectively reduce the computational cost during the training of multibranch structures [21]. Fig. 3(a) depicts the organization of MVCF, which includes three distinct branches. Given the input features $F_O = F_4 \in \mathbb{R}^{B\times C\times H\times W}$, where B indicates the batch size, the output features F_M are calculated as

$$F_M = f_1(F_O) + f_2(F_O) + f_3(F_O) \tag{4}$$

where $f_1(\bullet)$, $f_2(\bullet)$, and $f_3(\bullet)$ denotes the mapping functions of branches ①, ②, and ③ in Fig. 3(a). In the inference stage, structural reparameterization is employed to accomplish the weight fusion of the three branches. The specific steps are listed below.

Step 1: Fusion of branch ①. To save the graphic memory, bias terms are usually excluded from the convolutional layer before the batch normalization (BN) layer, so $DW\text{Conv}_{3\times3}(F_O) = W^{(3)}(F_O)$, where $W^{(3)}$ denotes the weight of $DW\text{Conv}_{3\times3}$. The corresponding branch ① is then computed and rearranged as follows:

$$\frac{\gamma^{(3)}W^{(3)}(F_O)}{\sqrt{\sigma^2+\varepsilon}} + \left(\beta^{(3)} - \frac{\gamma^{(3)}\mu}{\sqrt{\sigma^2+\varepsilon}}\right) = W_f^{(3)}(F_O) + b_f^{(3)} \tag{5}$$

where $\gamma^{(3)}$ and $\beta^{(3)}$ are the learnable parameters of the BN layer, μ and σ represent the mean and standard deviation of the current batch, ε is the minimum value term, and the first and second terms on the left in (5) are the fused weights $W_f^{(3)}$ and bias $b_f^{(3)}$.

Step 2: Fusion of branch ②. This branch fusion method involves padding 1×1 kernels with zeros to create a new 3×73 kernel, followed by weight fusion using the step from branch ①. Hence, the output of branch ② after conversion is

$$f_2'(F_O) = \mathcal{B}\left(f_{DW\text{Conv}_{1\times1\to3\times3}}(F_O)\right) = W_f^{(1)}(F_O) + b_f^{(1)} \tag{6}$$

where $\mathcal{B}$ represents the BN layer.

Step 3: Fusion of branch ③. The identify branch is equivalent to a degraded 1×71 DWConv with a kernel value of one. Thus, the weights can be successively fused using the steps of branches ② and ①. The final single-branch DWConv weights for the inference stage in Fig. 3(a) are derived by summing the corresponding fusion weights obtained in steps 1, 2, and 3, which is denoted as

$$F_M = \tilde{f}(F_O) = \Delta((\tilde{W}_f)(F_O) + \tilde{b}_f) \tag{7}$$

where Δ denotes the sigmoid linear unit (SiLU) activation function; $\tilde{W}_f = W_f^{(3)} + W_f^{(1)} + W_f^{(0)}$ and $\tilde{b}_f = b_f^{(3)} + b_f^{(1)} + b_f^{(0)}$ represent the weights and bias of a single branch after fusion.

2) Efficient Cross-Feature Interaction (ECFI): ECFI is designed to enhance the fusion of critical defect features in the compressed and concatenated MFL signals, and it not only enriches the gradient flow but also reduces the computational overhead [22], [23]. Fig. 3(b) illustrates the ECFI structure. Given the feature $F_M \in \mathbb{R}^{C\times H\times W}$ produced by MVCF as input, first, the feature F_{M1} and F_{M2} are obtained after Conv#1 and Conv#2, respectively, where $F_{M1}, F_{M2} \in \mathbb{R}^{C/2\times H\times W}$. Then,

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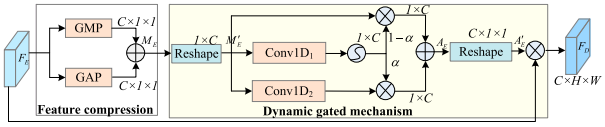


Fig. 4. Architecture of DGCA.

F'_{M1} is generated from F_{M1} using stacked N residual blocks, which is computed as

$$F'_{M1} = \prod_{i=1}^{N} (F_{M1}^{i-1} + f_{GConv1}^{i}(f_{GConv0}^{i}(F_{M1}^{i-1}))) \tag{8}$$

where $\prod_{i=1}^{N}(\bullet)$ represents the stack of N residual blocks, and GConv signifies ghost convolution. The final ECFI output F_E is acquired as follows:

$$F_E = f_{\text{Conv}\#3}((F_{M2}) \parallel (F'_{M1})) \tag{9}$$

where $\parallel$ denotes concatenation in the channel dimension.

3) Dynamic Gated Channel Attention (DGCA): Since the DWConv structure used in MVCF and ECFI lacks effective inter-channel communication, the DGCA module is designed to promote inter-channel competition and cooperation to deeply explore the abstract representation of defects, as presented in Fig. 4. To focus on the salient and universal channels of feature F_E, feature compression is performed simultaneously using global max pooling (GMP) and global average pooling (GAP) to obtain the channel weights M_E, computed as

$$M_E = \left(\max + \frac{1}{HW}\right) \sum_{i=1}^{H} \sum_{j=1}^{W} F_E(i,j). \tag{10}$$

For lightweight design, this article transforms $M_E \in \mathbb{R}^{C\times 1\times 1}$ into $M'_E \in \mathbb{R}^{1\times C}$ and uses a simple 1-D convolution for crucial channel selection. Meanwhile, the dynamic gated mechanism is constructed by combining representative channel information after convolutional aggregation with the sigmoid activation function, which adaptively guides the flow of critical information. Finally, the channel attention weight A_E is gathered and computed as follows:

$$A_E = M'_E \otimes (1-\alpha) + f_{\text{Conv1}D_2}(M'_E) \otimes (\alpha) \tag{11}$$

where α equals $\eta(f_{\text{Conv1}D_1}(M'_E))$ and η denotes the sigmoid activation function. After reshaping A_E into A'_E, the final feature output of DGCA is computed as $F_D = A'_E \otimes F_E$.

Table I details the efficient feature encoder's five phases, which are meant to assist with the progressive extraction of discriminative features linked to weld defects. Specifically, phase 1 comprises a 6×76 large kernel convolutional layer to expand the model's receptive field and capture finer-grained information inside the image. Phases 2 and 3 are built by stacking MVCF and ECFI modules to improve nonlinear learning capacity and effectively learn the intrinsic distribution of weld defects. To enhance the model's extraction of high-dimensional semantic information, the DGCA module is incorporated into phases 4 and 5, encouraging inter-channel feature interaction.

TABLE I
CONFIGURATION OF THE EFFICIENT FEATURE ENCODER

Name	Configuration	Parameters	Output shape
Phase 1	Conv(32,6,2,2)[1]	3520	$F_1: 32 \times 7320 \times 7320$
Phase 2	MVCF(64,3,2,0) + ECFI(64)	10 552	$F_2: 64 \times 7160 \times 7160$
Phase 3	MVCF(128,3,2,0) + 2×7ECFI(128)[2]	41 952	$F_3: 128 \times 780 \times 780$
Phase 4	MVCF(256,3,2,0) + 3×7 ECFI(256) + DGCA(256)	168 618	$F_4: 256 \times 740 \times 740$
Phase 5	MVCF(512,3,2,0) + ECFI(512) + DGCA(512) + SPPF(512)	1 228 726	$F_5: 512 \times 720 \times 720$
Total	-	1 453 368	-

[1]Where (32,6,2,2) represent the output channel, kernel size, stride, and padding, respectively.
[2]Where 2× refers to the stacking frequency of residual blocks within ECFI.

D. Dual-Path Feature Aggregation

As depicted in Fig. 3, a dual-path feature aggregation network is constructed by integrating three lightweight modules to improve the recognizability of defective features. Considering the salience of the features, multiscale feature aggregation is performed using the feature maps $\{F_3, F_4, F_5\}$ obtained from the last three stages of the efficient feature encoder. To enhance semantic representation, the deep abstract features are first transmitted to the shallow layer via a top-down subnet, which is denoted as

$$\begin{cases} M_5 = f_{\text{conv}}(F_5) \\ M_4 = f_{\text{conv}}[f_{ED}(\partial_{\uparrow}^{2}(M_5) \parallel F_4)] \\ M_3 = f_{\text{ED}}[(\partial_{\uparrow}^{2}(M_4)) \parallel F_3] \end{cases} \tag{12}$$

where f_{conv} represents the adaptive convolutional layer, f_{ED} denotes the integrated mapping function of ECFI and DGCA, and $\partial_{\uparrow}^{2}$ represents the upsampling operation. By employing the top-down subnet, features $\{M_3, M_4, M_5\}$ with rich semantics can be obtained. Subsequently, the bottom-up subnet is employed to propagate shallow-level localization information to deep layers in order to enhance localization capabilities at multiple scales. The calculation proceeds as follows:

$$\begin{cases} F'_3 = M_3 \\ F'_4 = f_{\text{ED}}[f_M(F'_3) \parallel M_4] \\ F'_5 = f_{\text{ED}}[f_M(F'_4) \parallel M_5]. \end{cases} \tag{13}$$

The accurate and rapid detection of weld defects can be ensured based on the finally obtained enhanced feature $\{F'_3, F'_4, F'_5\}$.

E. Refined Rotated Boundary Decision

Complex contours of weld defects in MFL signals such as slender shapes with angles pose challenges to distinguishable feature learning, so more accurate regularization algorithms are needed to constrain the model for fine optimization. To this end, this study presents a refined rotated boundary decision method that includes a fast decoupled head (FD-Head) and three regularization criteria. In comparison to the coupled head, the decoupled head excels at capturing texture features for classification and edge features for localization. However, using the decoupled head increases the computational complexity. Therefore, in this article, an innovative FD-Head with MVCF

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is proposed to cost-effectively enhance the feature learning of weld defects. In addition, three regularization criteria are established to constrain the model towards targeted gradient updating for meticulous parameter optimization: classification, DC, and AR loss. Among them, the classification loss is achieved by using binary cross entropy (BCE) to prevent different classes having the same semantic associations, denoted as

$$\mathcal{L}_{cls} = -y_{i,j}^{cls} \log \eta\left(p_{i,j}^{cls}\right) - \left(1 - y_{i,j}^{cls}\right) \log\left(1 - \eta\left(p_{i,j}^{cls}\right)\right) \tag{14}$$

where $p_{i,j}^{cls}$ signifies the predicted category probability for the i-th anchor in the j-th grid, and $y_{i,j}^{cls}$ denotes the true category label associated with that anchor.

Since the traditional rotated bounding box (RBB) loss encounters boundary discontinuities and square-like issues, this study converts the RBB to a two-dimensional Gaussian distribution to improve the robustness of the prediction [18]. Given the label $Y_i{}^r = (x_i, y_i, w_i, h_i, \theta_i)$ of the i-th RBB in the j-th MFL image, it can be transformed into a Gaussian distribution represented by $Y_i{}^r \sim \aleph_r\left(\kappa_r, \mathbf{\Psi_r}\right)$ using a rotation transformation matrix. Here, $\kappa_r = (x_i, y_i)^{\mathcal{T}}$ is the centroid of the RBB and $\mathbf{\Psi_r}$ is the covariance matrix structured as

$$\mathbf{\Psi_r}^{\frac{1}{2}} = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix} \begin{pmatrix} \frac{w_i}{2} & 0 \\ 0 & \frac{h_i}{2} \end{pmatrix} \begin{pmatrix} \cos\theta_i & \sin\theta_i \\ -\sin\theta_i & \cos\theta_i \end{pmatrix}. \tag{15}$$

Assuming that the predicted RBB follows a Gaussian distribution $Y_i^p \sim \aleph_p(\kappa_p, \mathbf{\Psi_p})$, the Kullback–Leibler Divergence (KLD) between the predicted and actual RBB can be expressed as

$$D_{kl}\left(\aleph_p \| \aleph_r\right) = \frac{1}{2}(\kappa_p - \kappa_r)^{\mathcal{T}} \mathbf{\Psi_r^{-1}} (\kappa_p - \kappa_r) + \frac{1}{2}\mathbf{Tr}\left(\mathbf{\Psi_r^{-1}\Psi_p}\right) + \frac{1}{2} In \left|\frac{\mathbf{\Psi_r}}{\mathbf{\Psi_p}}\right| - 1 \tag{16}$$

where $\mathbf{Tr}$ represents the matrix's trace. The DC loss $\mathcal{L}_{\text{dc}}$ is thus expressed as

$$\mathcal{L}_{\text{dc}} = 1 - \frac{1}{\tau + \log\left(D_{kl} + 1\right)} \tag{17}$$

where τ denotes the modulation parameter, which is set to 1.

The AR loss, inspired by [24], aims to direct the model to focus on the weld's slender features. First, the aspect ratio v between the real and predicted RBB is calculated as

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w_i}{h_i} - \arctan\frac{\hat{w}_i}{\hat{h}_i}\right)^2. \tag{18}$$

Therefore, the AR loss $\mathcal{L}_{\text{ar}}$ is denoted as follows:

$$\mathcal{L}_{\text{ar}} = \frac{v^2}{1 + v - \text{IoU} + \text{eps}} \tag{19}$$

where IoU signifies the intersection over union between the predicted RBB and the ground truth. Thus, the total loss $\mathcal{L}$ of LRNet is

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \gamma_{\text{dc}}\mathcal{L}_{\text{dc}} + \beta_{\text{ar}}\mathcal{L}_{\text{ar}} \tag{20}$$

where λ_{cls}, γ_{dc}, and β_{ar} are the balance coefficients.

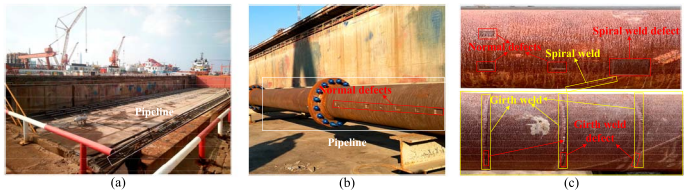


Fig. 5. Pipeline experimental platform in north China. (a) Experimental site. (b) Experimental pipeline. (c) Actual weld defects.

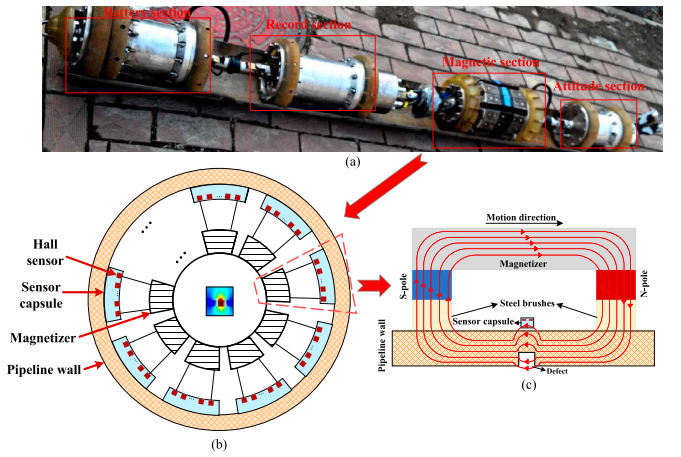


Fig. 6. MFL inspection robot. (a) Physical robot. (b) Cross section of the magnetic section. (c) Longitudinal section of the magnetic section.

IV. Experiments and Discussion

To validate the effectiveness of our approach, a series of experiments are conducted on MFL data from the experimental site (i.e., MFL-Experiment). Specifically, comprehensive performance and lightweight comparisons are performed using advanced methods such as horizontal bounding box, rotated bounding box, and instance segmentation. In addition, a thorough generalization analysis using MFL data from real-world pipelines (i.e., MFL-Real-world) is performed to evaluate the practical engineering value of the proposed methods in real-world scenarios. Finally, ablation experiments are executed to evaluate the individual contributions of different components within our framework.

A. Experimental Settings

1) Sample Generation: The raw signals are acquired from a pipeline experimental platform in northern China, as depicted in Fig. 5. The tested pipeline is made of X65 carbon steel, with a length of approximately 100 m, an outer diameter of 325 mm, and a wall thickness of 12.7 mm. The MFL signal is obtained through the Hall sensor carried by the MFL inspection robot, as shown in Fig. 6(a). Fig. 6(b) and 6(c) illustrate the cross and longitudinal sections of the magnetic section. As shown in Fig. 6(b), the Hall sensors are uniformly distributed in sensor capsules around the magnetic section, with each capsule containing 6 axial sensors for a total of 288 axial signals. The MFL inspection mechanism is depicted in Fig. 6(c), where the

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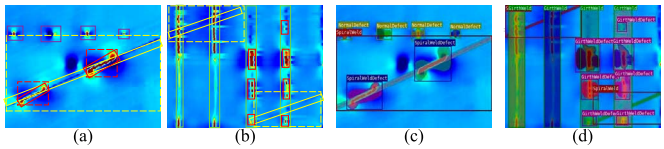


Fig. 7. Typical samples of weld defects. (a) and (b) are annotated with horizontal bounding boxes and rotated bounding boxes. (c) and (d) are annotated with masks.

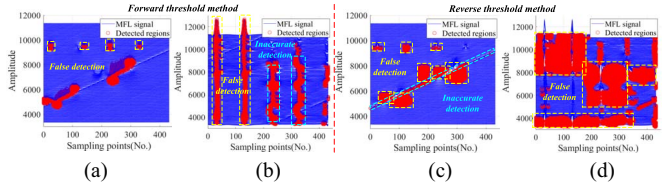


Fig. 8. Detection results of traditional threshold-based methods. (a), (b) and (c), (d) show the detection results of forward and reverse threshold methods.

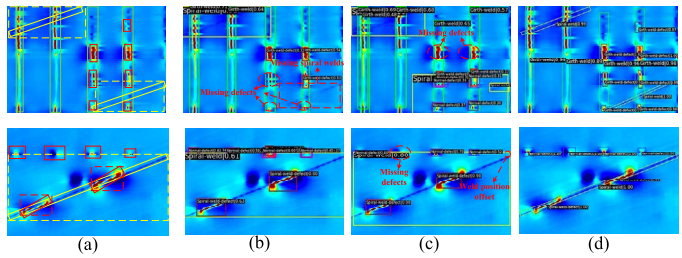


Fig. 9. Detection results of MFL weld defect datasets with different annotation forms. (a) Ground truth. (b) LED-Net. (c) LF-YOLO. (d) LRNet.

presence of defects in the pipe wall generates a magnetic field leakage that is captured by the carried sensor capsule for signal acquisition. In practice, the sampling interval is set to 0.002 m, so the size of the collected axial signal is $50\,000 \times 7288$. According to the data processing method mentioned in Section III-B, a total of 841 image samples are generated, which are mainly categorized into welds and defects, as shown in Fig. 5(b) and 5(c). Specifically, welds can be further divided into spiral welds (SWs) and girth welds (GWs). Additionally, defects can be further divided into pipe wall defects (normal defects, NDs) and weld defects (spiral weld defects, SWDs, and girth weld defects, GWDs).

To thoroughly investigate the effectiveness of our method, we carefully annotated defective samples using horizontal bounding boxes (HBB), RBB, and masks. Notably, instance masks within the same category are not distinguished for ease of engineering application. Specifically, the RBB annotation is represented as (x, y, w, h, θ), where θ is the angle between w and the x-axis, with a range of $[0, \pi]$. In practical training, we converted (x, y, w, h, θ) into sequential quadrilateral annotations, i.e., $(x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4)$. Fig. 7 illustrates typical examples of weld defect annotations.

2) Implementation Details: To accelerate convergence, common objects in context (COCO) pre-trained weights are used to initialize the model. The model is trained for 200 epochs with a batch size of 8, with the first three epochs used for warming up. Stochastic gradient descent (SGD) with momentum is employed for parameter optimization, with a momentum of 0.779 and a decay rate of 0.00058. The initial learning rate is set to 0.17 and is decayed using a cosine annealing strategy. To enrich the diversity of the training samples, data augmentation strategies such as left-right and top-down flips are introduced. Additionally, the division ratio of training and testing images is 8:2. The size of the training and testing images is scaled to 640×7640, and the balance coefficients for the classification loss, DC loss, and AR loss are set to 0.299, 0.0539, and 0.011, respectively. All code is developed in the PyTorch framework, and all experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 3070 GPU and an Intel Core i7-12700KF CPU.

3) Evaluation Metrics: The evaluation considered two factors: performance and lightweight. For performance evaluation, we used mainstream metrics employed in defect detection, including average precision (AP), mean average precision (mAP), average recall (AR), mean average recall (mAR), false positive rate (FPR), and mean false positive rate (mFPR). AP, AR, and FPR are used to measure the precision, recall rate, and false detection rate of the model for single-class predictions. Correspondingly, mAP, mAR, and mFPR reflect the model's overall performance in these three properties. It is noteworthy that all metrics are calculated with an IoU of 0.5. Additionally, to comprehensively assess the lightweight, we used parameters (Params), floating-point operations (FLOPs), and post-training model weights to measure the model's complexity. We also used training time and frames per second (FPS) to assess the actual computational burden. It is noteworthy that the training time is calculated as the execution time per epoch of the model.

B. Experiment 1: Comparison of Performance

To comprehensively assess the efficacy of our method, we select traditional threshold-based method [28], state-of-the-art (SOTA) HBB-based detection methods such as LED-Net [12] and LF-YOLO [10], as well as RBB-based detection approaches including rotation-equivariant detector (ReDet) [25], KLD [18], single-shot alignment network (S2A-Net) [26], and oriented region-CNN (OR-CNN) [27], for performance and lightweight comparison. Furthermore, since instance segmentation is a fine-grained pixel-level detection tool, advanced instance segmentation methods like Mask-RCNN [29], Queryinst [30], and Instaboost [31] are also used for comparative evaluations.

1) Comparison With Threshold-Based Detection Methods: Based on the MFL signal characterization, in order to completely detect all useful regions with the thresholding method, we selected a bidirectional thresholding method [28] that includes forward and reverse thresholding, which aims to detect high-amplitude (amplitude $>$ threshold) and low-amplitude (amplitude $<$ threshold) signals, and the results are shown in Fig. 8. From Fig. 8, despite the threshold adjustment by manual intervention, the detection results are still unsatisfactory, with

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TABLE II
EXPERIMENTAL RESULTS OF PERFORMANCE COMPARISON OF DETECTION METHODS BASED ON HBB AND RBB ON MFL-EXPERIMENT

Methods	Annotation	AP					mAP	FPR					mFPR	AR					mAR
		SW	GW	SWD	GWD	ND		SW	GW	SWD	GWD	ND		SW	GW	SWD	GWD	ND	
LED-Net [12]	HBB	81.5%	85.1%	85.7%	81.0%	83.0%	83.3%	8.7%	7.6%	9.1%	13.4%	7.5%	9.3%	82.3%	79.3%	87.3%	84.8%	80.0%	82.7%
LF-YOLO [10]	HBB	82.6%	85.6%	86.1%	82.3%	84.7%	84.3%	9.1%	9.2%	8.0%	6.2%	5.3%	7.6%	80.5%	82.6%	85.3%	85.1%	82.1%	83.1%
ReDet [25]	RBB	89.5%	87.9%	90.4%	89.5%	90.5%	89.6%	3.7%	5.8%	3.9%	7.8%	5.7%	5.4%	87.5%	84.2%	92.9%	85.7%	91.2%	88.3%
KLD [18]	RBB	89.7%	88.0%	**94.6%**	93.6%	92.5%	91.7%	4.6%	3.2%	3.4%	5.7%	4.5%	4.3%	87.6%	88.7%	93.1%	91.2%	87.4%	89.6%
S2A-Net [26]	RBB	88.9%	89.7%	93.5%	94.1%	93.8%	92.0%	4.0%	4.4%	3.7%	4.2%	2.1%	3.7%	91.8%	90.3%	91.0%	92.2%	94.0%	91.9%
OR-CNN [27]	RBB	93.2%	93.7%	93.8%	93.6%	93.0%	93.5%	**1.8%**	3.6%	2.5%	5.3%	3.6%	3.4%	90.3%	**93.8%**	89.5%	91.4%	90.0%	91.0%
LRNet (Ours)	RBB	**93.8%**	**94.2%**	94.3%	**94.1%**	**93.9%**	**94.1%**	2.5%	**2.0%**	**1.8%**	**3.8%**	**1.5%**	**2.3%**	**92.4%**	93.5%	**93.8%**	**94.6%**	**95.5%**	**94.0%**

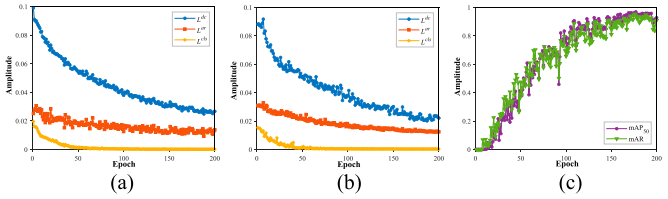


Fig. 10. Changes in the loss and accuracy of LRNet during the training process. (a) Training loss. (b) Validation loss. (c) mAP and mAR.

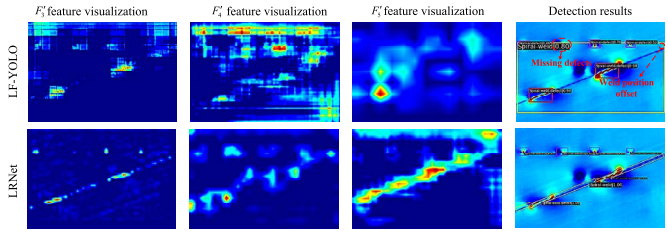


Fig. 11. Visualization of aggregated features for LF-YOLO and LRNet.

a large number of false and inaccurate detections. We argue that due to the strong amplitude coupling between welds and defects, setting a fine threshold to distinguish them is arduous and cumbersome, which seriously hinders practical engineering applications. On the contrary, in Fig. 9(d), the proposed LRNet significantly improves the performance and provides accurate detection results. This indicates that the LRNet can distinguish strong coupling to accurately regress the boundaries of welds and defects, which is friendly for practical engineering applications.

2) Comparison With HBB-Based Detection Methods: According to the results in Table II, LRNet outperforms the SOTA HBB-based detection methods. For instance, LRNet outperforms LF-YOLO by 9.8% and 10.9% in mAP and mAR, respectively, while reducing mFPR by 5.3%. This improvement demonstrates that the rotation-based detection method can suppress the interference of complex backgrounds and guide the model to focus on the key features of weld defects, thus reducing false and missed detections. Moreover, as depicted in Fig. 9, unlike the HBB-based detection method, the weld defect contours regressed by LRNet are highly consistent with the ground truth, and there are no missed or false detections in the test samples. Additionally, in Fig. 10, we present the changes in loss and accuracy during the training process, where it can be observed that, with model iterations, the training and validation losses exhibit an exponential decreasing trend, while mAP and mAR follow a logarithmic growth trajectory. This indicates that LRNet can capture the discriminative distribution of welds and defects to accurately predict the irregular contours of weld defects during training, thus demonstrating good fitting and generalization.

To comprehend the mechanism behind achieving high precision, we use a typical SWD sample to display features F_3', F_4', and F_5' in LF-YOLO and LRNet using class activation mapping

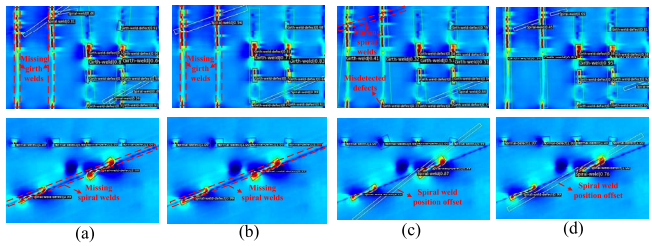


Fig. 12. Comparison of detection results based on different rotated detection methods. (a) ReDet. (b) KLD. (c) S2A-Net. (d) OR-CNN.

heatmaps, as shown in Fig. 11. In Fig. 11, LRNet has a more concentrated and distinct activation region in the shallow feature F_3' than LF-YOLO, especially for smaller NDs and SWDs. Furthermore, LRNet improves the activation response of larger weld and defect features in the deeper features F_4' and F_5', with a greater emphasis on slender features. The above study verifies that LRNet enables the model to focus on the macro- and micro-discriminative features of weld defects, thereby clarifying the coupled relation between welds and defects to improve the overall performance of the model.

3) Comparison With RBB-Based Detection Methods: Based on the results in Table II, LRNet outperforms advanced RBB-based detection methods. For instance, LRNet surpasses OR-CNN by 0.6% and 3.0% in mAP and mAR, respectively, while decreasing mFPR by 1.1%. This indicates that LRNet's refined rotated boundary decision guides the detector to learn plentiful rotation-invariant information, thereby enhancing the coarse-to-fine semantic representation of weld defects. Fig. 12 shows the visual detection results of the RBB-based approach. Unfortunately, compared with LRNet in Fig. 9(d),

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TABLE III
EXPERIMENTAL RESULTS OF PERFORMANCE COMPARISON WITH ADVANCED INSTANCE SEGMENTATION METHODS ON MFL-EXPERIMENT

Methods	Annotation	AP					mAP	mAR	mFPR
		SW	GW	SWD	GWD	ND			
Mask-RCNN [29]	Mask	92.6%	94.5%	93.4%	93.6%	92.8%	93.4%	92.6%	2.6%
QueryInst [30]	Mask	93.6%	**95.4%**	85.4%	86.6%	83.6%	88.9%	90.4%	3.4%
Instaboost [31]	Mask	93.5%	94.6%	94.0%	94.1%	92.7%	93.8%	93.4%	2.5%
LRNet (Ours)	RBB	**93.8%**	94.2%	**94.3%**	**94.1%**	**93.9%**	**94.1%**	**94.0%**	**2.3%**

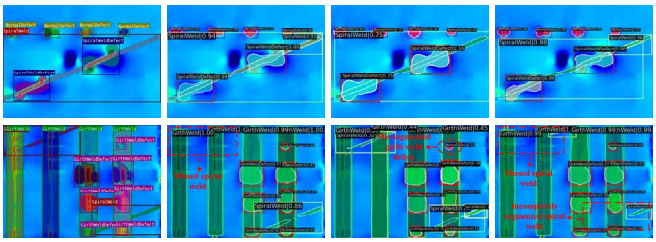


Fig. 13. Visualization results for instance segmentation. (a) Ground truth. (b) Mask-RCNN. (c) QueryInst. (d) Instaboost.

most RBB-based methods exhibit partial misses and false detections when detecting GWD and GW, and show varying degrees of offset when detecting SW. This further indicates the ability of LRNet to pinpoint defects and ensure pipeline safety.

4) Comparison With Instance Segmentation Methods: The instance segmentation methods were meticulously trained to ensure optimal performance. Statistical measurement results are provided in Table III, and visualization results are depicted in Fig. 13. According to Table III, both Mask R-CNN and Instaboost perform well in segmentation, with only a slight decrease of 0.7% and 0.3% in mAP compared with LRNet. However, as seen in Fig. 13, both methods failed to segment the crucial SW. Additionally, QueryInst incorrectly segmented the critical GWD, which is frustrating. These observed results suggest that inherent difficulties in pixel-level segmentation and the strong coupling between welds and defects may affect the promised performance.

C. Experiment 2: Comparison of Lightweight

To investigate LRNet's efficiency, we conducted a thorough lightweight analysis, and the results are reported in Table IV.

1) Comparison With HBB-Based Detection Methods: As can be seen from Table IV, compared with LF-YOLO with superior performance, LRNet reduces parameters, FLOPs, and weights by 0.6 M, 2.6 G, and 0.8 M, respectively. Additionally, LRNet's training time decreases by 6 s, while the inference speed increases by 6 FPS. This indicates that incorporating the idea of rotated detection improves LRNet's performance while maintaining comparable parameters. Although LRNet is slightly higher than LED-Net in model parameters and FLOPs, it outperforms LED-Net by 10.8% in mAP. Considering the industry's emphasis on striking a favorable balance between speed and accuracy, this gap is deemed acceptable.

TABLE IV
EXPERIMENTAL RESULTS OF LIGHTWEIGHT COMPARISON ON MFL-EXPERIMENT

Methods	Mask-RCNN	Queryinst	Instaboost	ReDet	KLD	OR-CNN	S2A-Net	LED-Net	LF-YOLO	LRNet
Params (M)	43.8	172.2	43.8	29.9	19.7	28.1	22.1	2.9	4.2	3.6
FLOPs (G)	245.4	464.5	245.4	22.4	62.2	57.1	56.6	7.0	10.2	7.6
Weight (M)	335.0	1973.0	335.0	232.0	152.0	215.0	169.0	6.8	8.1	7.3
Time (s)	107.0	335.0	104.0	56.0	27.0	32.0	32.0	18.0	25.0	19.0
FPS	18.0	14.0	18.0	27.0	46.0	44.0	44.0	84.0	73.0	79.0

2) Comparison With RBB-based Detection Methods: Table IV demonstrates that rotated detectors generally have larger model sizes and slower speeds. For instance, compared with OR-CNN, LRNet reduces parameters, FLOPs, and training time by 24.5 M, 49.5 G, and 13 s, respectively, while increasing the inference speed by 35 FPS. The results show that the lightweight design of LRNet can minimize the computational cost while maintaining accuracy, which makes it suitable for deployment in IPCs to complete weld defect detection online.

3) Comparison With Instance Segmentation Methods: We also assessed the complexity and speed of instance segmentation methods, as shown in Table IV. Table IV indicates that these segmentation-based methods exhibit extremely high model complexity and severely limited inference speeds, which suggests that the challenging nature of pixel-level classification requires Unfortunately, IPCs, as industrial devices with limited computational power, face limitations in deploying such complex models. Therefore, investigating lightweight segmentation methods for industrial applications would be an interesting direction.

D. Experiment 3: Comparison of Generalizability in MFL-Real-World Samples

1) Dataset: To thoroughly investigate the generalizability of the proposed method, we conducted an experimental analysis using MFL samples from a real undersea pipeline (i.e., MFL-real-world). MFL-Real-world contains 974 images, mainly divided into two categories: girth welds (GW) and defects (DE). Due to the uncertainties in the real world, the sample includes a large number of small and dense irregular corrosion defects, which greatly increase the challenge of detection. During the training process, we divide the ratio of training and validation images into 8:2 and use both HBB and RBB for detailed annotations.

2) Comparison of Performance: Table V lists the performance comparison test results on real-world samples. Due to the complexity of real-world samples, all methods inevitably experience performance degradation, but the proposed method still provides the best performance gains. For instance, compared with the SOTA HBB-based method (i.e., LF-YOLO), LRNet improves by 11.5% and 11.4% in mAP and mAR respectively, and reduces by 11.8% in mFPR. Additionally, in Fig. 14, we present two typical small defect detection results and heatmaps in the real world, showing that the minimum defect size that can be detected by LRNet is 12×710 (pixels) and its actual size is 5×75 (mm). Moreover, Fig. 14(b) and 14(c) demonstrate that LRNet can effectively activate small defect areas in shallow

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TABLE V
EXPERIMENTAL RESULTS OF PERFORMANCE COMPARISONS USING MFL-REAL-WORLD SAMPLES

Methods	Annotation	AP		mAP	FPR		mFPR	AR		mAR
		DE	GW		DE	GW		DE	GW	
LED-Net [12]	HBB	55.4%	81.5%	68.5%	26.8%	22.0%	24.4%	53.1%	76.3%	64.7%
LF-YOLO [10]	HBB	57.5%	82.3%	69.9%	22.3%	19.4%	20.9%	59.6%	85.3%	72.5%
ReDet [25]	RBB	61.3%	85.4%	73.4%	17.6%	14.7%	16.2%	65.4%	83.2%	74.3%
KLD [18]	RBB	60.2%	**91.8%**	76.0%	14.1%	15.8%	15.0%	67.8%	84.5%	76.2%
S2A-Net [26]	RBB	68.2%	88.9%	78.6%	9.8%	13.6%	11.7%	74.5%	**89.4%**	82.0%
OR-CNN [27]	RBB	69.1%	90.5%	79.8%	12.4%	9.3%	10.9%	77.8%	87.4%	82.6%
LRNet (Ours)	RBB	**71.5%**	91.2%	**81.4%**	**9.5%**	**8.6%**	**9.1%**	**79.2%**	88.7%	**83.9%**

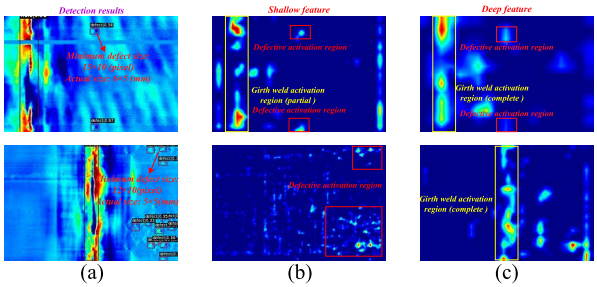


Fig. 14. Visualization of detection results and heat maps for MFL-Real-world samples. (a) Detection results. (b) and (c) are heat maps of shallow and deep features, respectively.

TABLE VI
EXPERIMENTAL RESULTS OF LIGHTWEIGHT COMPARISONS USING MFL-REAL-WORLD SAMPLES IN AN IPC ENVIRONMENT

Methods	ReDet	KLD	OR-CNN	S2A-Net	LED-Net	LF-YOLO	Ours(LRNet)
Params (M)	29.9	19.7	28.1	22.1	2.9	4.2	3.6
FLOPs (G)	22.4	62.2	57.1	56.6	7.0	10.2	7.6
FPS	2.0	6.0	4.0	4.0	16.0	11.0	15.0

features and focus on the slender features of girth welds in deep features. These observations fully illustrate that LRNet can extract distinguishable features of welds and defects, enabling it to detect smaller defects under the influence of complex welds.

3) Comparison of Lightweight: For online validation in industrial sites, we utilize an IPC with limited computational resources for lightweight comparison experiments, as reported in Table VI. In Table VI, LRNet provides competitive inference speeds, e.g., LRNet improves the inference speed by 11 FPS compared with OR-CNN, and 4 FPS compared with LF-YOLO. These results fully demonstrate that the lightweight design of LRNet effectively compresses the model complexity and improves the inference speed, which makes it suitable to be deployed in resource-constrained IPCs in industrial sites and accomplish online detection.

E. Experiment 4: Ablation Analysis

In this section, we conduct ablation experiments to evaluate the efficacy of three components in the overall framework: the efficient feature encoder (EFE), dual-path feature aggregation (DPFA), and refined rotated boundary decision (RRBD). The experimental results are presented in Table VII.

TABLE VII
ABLATION EXPERIMENTS FOR COMPONENT DESIGN EFFECTIVENESS

Methods	Training		Inference		FPS	mAP
	Params (M)	FLOPs (G)	Params (M)	FLOPs (G)		
Base	7.0	16.0	7.0	15.8	61.0	89.5
+EFE	4.3	8.9	4.3	8.7	80.0	88.3
+DPFA	3.4	7.0	3.3	6.7	83.0	90.2
+RRBD	3.6	7.9	3.6	7.6	79.0	94.1

1) Effectiveness of EFE: Table VII indicates that after using EFE, the proposed method reduces 2.7M parameters and 7.1G FLOPs with only 1.2% loss of mAP, and improves the inference speed by 19 FPS. This result proves that the lightweight design of EFE can effectively reduce the model's computational burden and improve its ability to extract identifiable features, which is beneficial for deployment in IPCs.

2) Effectiveness of DPFA: The results in Table VII show that the addition of DPFA not only reduces the complexity of the model and slightly improves its speed, but also enhances its mAP by 1.9%. These results show that because the dual-path structure and the embedding of three modules greatly enrich the semantic and localization information of multiscale features, LRNet exhibits stronger detection accuracy of weld defects while compressing the computational cost.

3) Effectiveness of RRBD: In Table VII, the introduction of RRBD slightly increases the model complexity and reduces its inference speed while improving its mAP by 3.9%. This result shows that the designed FD-Head can effectively learn the inherent nonlinear relation between welds and defects. In addition, the proposed DC and AR losses can guide the model to focus on the salient features of defects.

V. CONCLUSION

In this article, an online intelligent detection method named LRNet is proposed for the real-time identification of weld defects. Specifically, efficient feature encoder and dual-path feature aggregation can extract and aggregate distinguishable representations of weld defects to clarify the coupled relations between welds and defects. Refined rotated boundary decision can guide the model to perform finer regression on weld defect contours. Additionally, the lightweight design of the model greatly reduces its complexity and computational cost.

Experimental validation shows that LRNet strikes a favorable balance between speed and accuracy. Compared with SOTA HBB-based models (e.g., LF-YOLO), LRNet outperforms it by 10% in mAP with comparable computational cost. Compared with robust RBB-based models (e.g., OR-CNN), LRNet improves the inference speed by 35 FPS and reduces the training time by 13 s without compromising the performance. Furthermore, it significantly reduces the computational cost compared with instance segmentation methods. These experimental results demonstrate that LRNet is a powerful lightweight industrial weld defect detector that can fulfill the real-time online inference requirements for IPCs. Additionally,

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generalization tests on real-world pipelines prove that LRNet can detect smaller defects affected by welds, demonstrating its strong generalization and robustness.

Due to the limited training examples, our future research will investigate image generation techniques or conduct on-site experiments to expand the dataset while deeply validating the model's generalization and transferability performance. Furthermore, considering that segmentation can finely delineate defect profiles but requires considerable parameters and complexity, we will also focus on designing an industrial-grade, lightweight semantic segmentation detection method.

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